

1.DATASET PREVIEW

...

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps: [Generate code with df](#) [New interactive sheet](#)

2.DATASET INFO

```
df.info()
```

...

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 200 entries, 0 to 199  
Data columns (total 5 columns):  
#   Column                Non-Null Count  Dtype  
---  ---                -  
0   CustomerID            200 non-null    int64  
1   Genre                  200 non-null    object  
2   Age                    200 non-null    int64  
3   Annual Income (k$)     200 non-null    int64  
4   Spending Score (1-100) 200 non-null    int64  
dtypes: int64(4), object(1)  
memory usage: 7.9+ KB
```

3.MISSING VALUES

```
df.isnull().sum()
```

...

	0
CustomerID	0
Genre	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0

dtype: int64

4.DATASET SHAPE

```
df.shape
```

```
(200, 5)
```

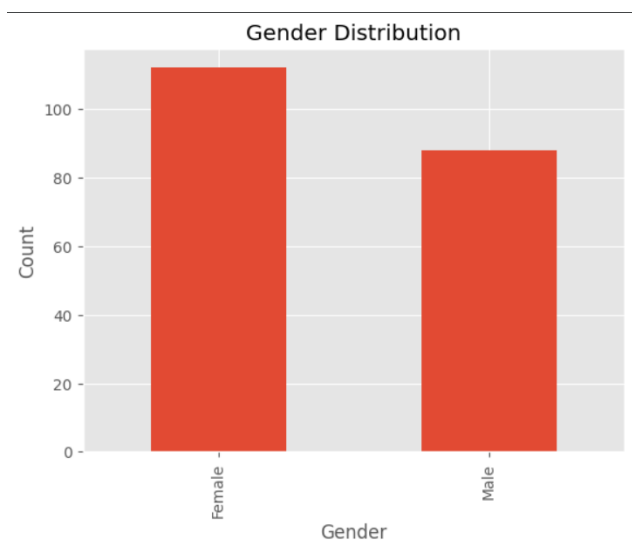
5. REMOVE DUPLICATES

```
df.drop_duplicates(inplace=True)
```

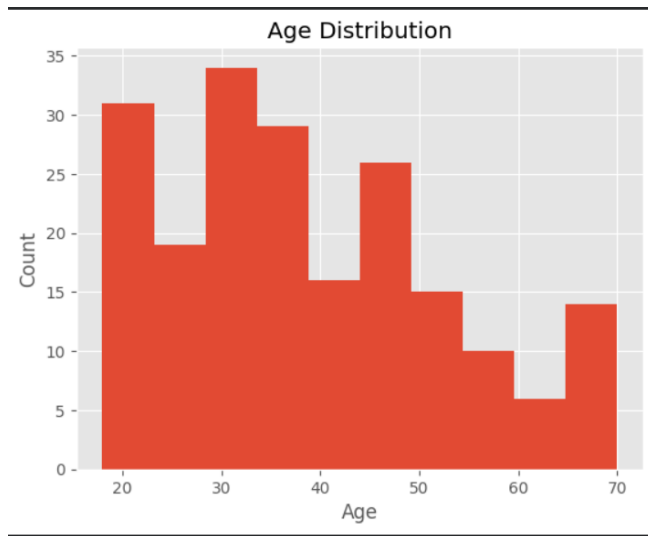
```
print(df.shape)
```

```
(200, 5)
```

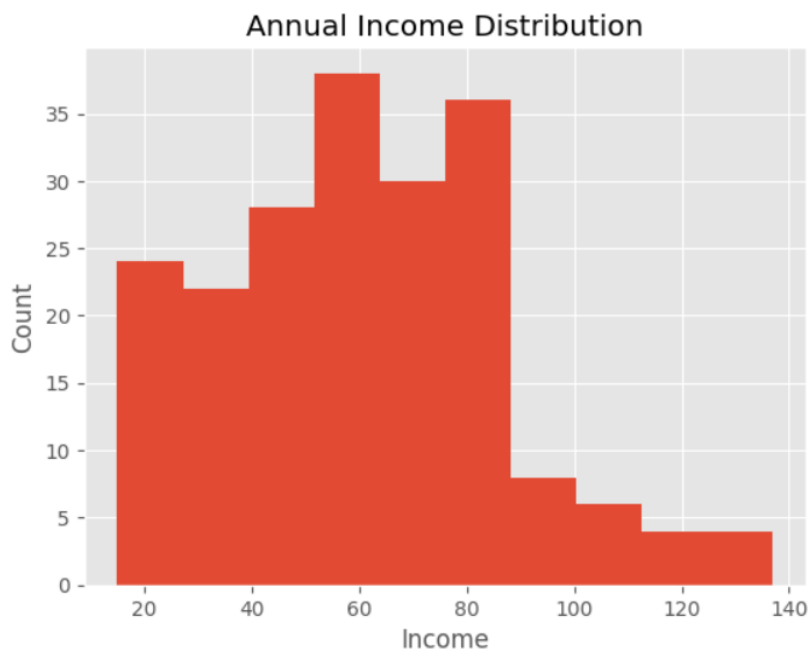
6.GENDER DISTRIBUTION



7.AGE DISTRIBUTION



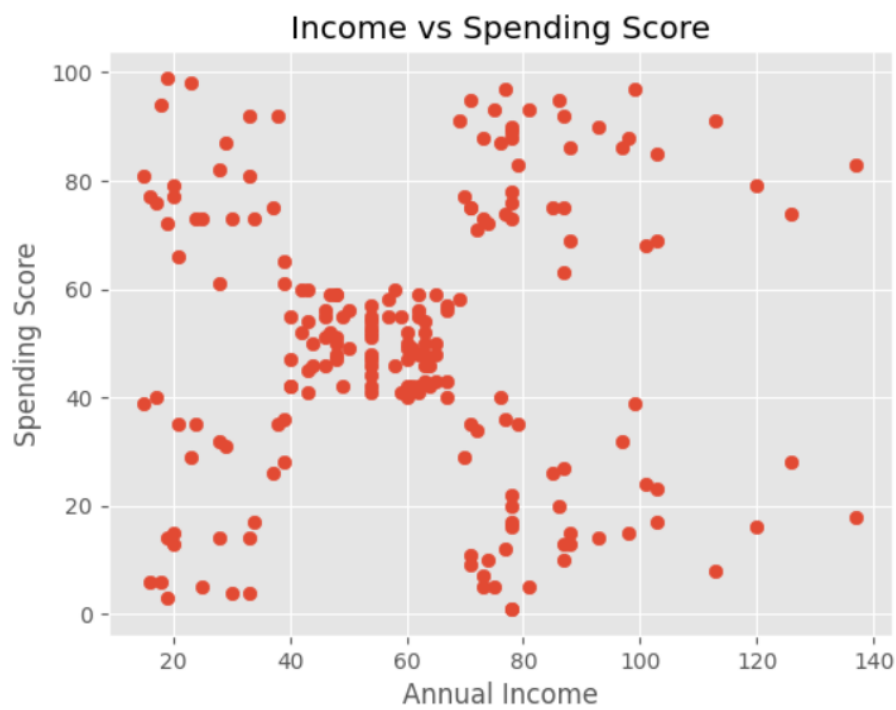
8.ANNUAL INCOME DISTRIBUTION



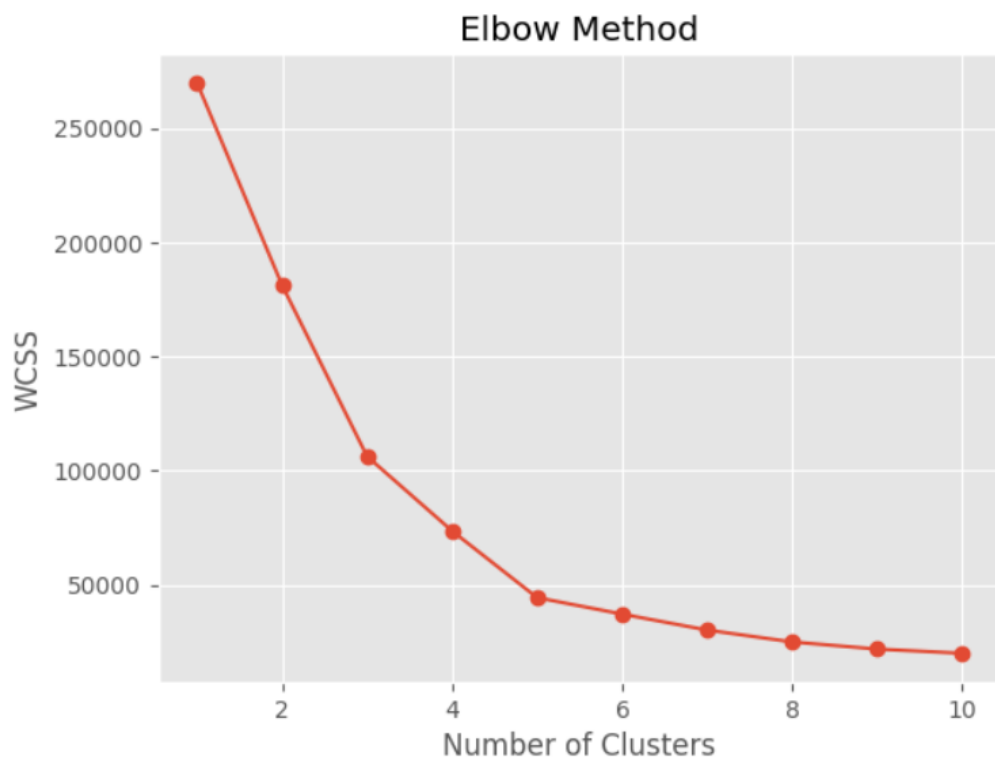
9.SPENDING SCORE DISTRIBUTION



10.SCATTER PLOT



11. ELBOW METHOD



12. CLUSTERED DATA

df.head()

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4

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13.CLUSTER SEGMENTS

